EEG-Based Binary Seizure Classification Under Clinical Realism: Imbalance, Artifacts, and Interpretability

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***Abstract*—The automated interpretation of continuous clinical electroencephalograms (EEG) remains a critical bottleneck in epilepsy diagnosis, largely due to extreme class imbalance, severe artifact contamination, and the tendency of deep learning models to exploit dataset-specific shortcuts. In this study, we present a comprehensive, multi-modal deep learning framework designed to explicitly overcome these clinical barriers. We first introduce a robust signal preprocessing pipeline utilizing subject-level Median and Interquartile Range (IQR) normalization, paired with a longitudinal bipolar montage to actively suppress amplitude distortion and spatial noise. Leveraging this foundation, we engineer a Heterogeneous Signal Ensemble that fuses orthogonal temporal (1D EEGNet) and spectral (2D CNN-LSTM) feature spaces. Fortified by a deterministic temporal continuity wrapper, this ensemble achieves a highly competitive 0.8108 F_1-macro and 0.9035 AUC-ROC on the strictly patient-disjoint TUH Seizure test split. Finally, to bridge the gap between predictive accuracy and clinical trust, we deploy a multi-tiered Explainable AI (XAI) framework. Through quantitative occlusion sensitivity and attribution mapping, we prove that our architectures autonomously suppress non-cerebral biological artifacts and successfully isolate genuine neurophysiological biomarkers. These findings establish a highly stable, clinically faithful pathway for deploying automated seizure detection in real-world diagnostic environments.**

I. Introduction

Epilepsy constitutes a major neurological disorder, affecting approximately 50 million people of a wide range of ages worldwide [1]. It is characterized by abnormal electrical activity causing different types of seizures or unusual behaviors, which can cause unexpected sensations and sometimes loss of awareness [1]. The risk of premature death in people with epilepsy is up to three times that of the general population [1].

Electroencephalography (EEG) is a noninvasive technique that is primarily used to capture the brain's activity via a sensor chain that consists of several sensors known as electrodes placed on the scalp to measure the electrical activity on the scalp [2] as illustrated in Fig. 1.

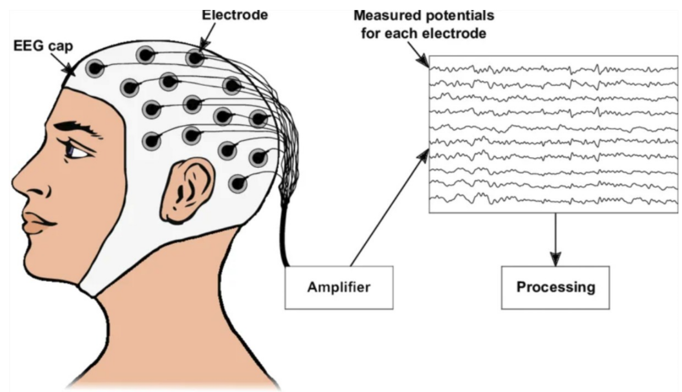

Fig. 1. EEG Sensor Array / Electrodes Placement.

Raw EEG signals contain rich temporal and spatial features that can be utilized and classified for the aim of seizure-related tasks [3]. Despite its central role in clinical practice, the interpretation of long-duration EEG recordings is labor-intensive [4], requires significant expertise, and is subject to inter-rater variability, motivating sustained interest in automated seizure detection and classification methods [5]. Furthermore, the visual interpretation of EEG is highly challenging; around 30% of patients are misdiagnosed with epilepsy due to the over-reading of benign EEG features [6]. Thus, these factors encourage motivation to enhance the current state of seizure detection with automatic and reliable alternatives.

Over the past few years, advances in machine learning and deep learning have led to a significant growth in automated multi-class classification EEG-based seizure detection systems. The availability of large-scale clinical datasets, most notably those derived from routine hospital recordings such as the Temple University Hospital EEG Corpus [7], has facilitated the development of increasingly complex models, including convolutional, recurrent, attention-based, hierarchical, and graph-structured architectures [8], [9]. As a result, advanced deep learning architectures, including attention-based deep CNN models, have achieved strong performance in this task [8]. Additionally, more recent work has demonstrated a level of trustworthiness utilizing Explainable Artificial Intelligence (XAI) techniques [10]. Certain methods have shown consistently reliable results in binary seizure classification regardless of the architecture, with minimal degradation when introducing explainability methods [11], contributing to a growing perception that seizure classification may be approaching a mature stage under modern deep learning frameworks [3].

However, this perception warrants careful reviewing alongside properly trusted evaluation when considered beside the fundamental properties of EEG signals. EEG does not provide a direct measurement of localized neural activity [2], [12]; instead, scalp-recorded potentials represent the collective, synchronized post-synaptic activity of significant neuronal populations, pyramidal cells for the most part [2], [12]. As signals propagate through brain tissue, skull, and scalp, volume conduction introduces inevitable spatial blurring, making the rendering task of exact source localization ill-posed [4]. Consequently, EEG offers high temporal fidelity at the expense of neglecting spatial aspects of the signals, a trade-off that cannot be avoided through algorithmic sophistication without additional physiological and biophysical constraints and multimodal information [13].

In addition to these intrinsic limitations, real-world clinical EEG recordings are extremely prone to contamination by different types of artifacts, including physiological sources such as eye blinks, muscle activity, and cardiac signals, as well as non-physiological sources including electrode pop and power-line interference [14], [15], [16]. Such artifacts often overlap with neural activity in both temporal and spectral domains and may exhibit amplitudes exceeding those of cortical signals [14], [15], [16]. Several studies and surveys in EEG-based brain-computer interfaces and clinical applications indicate that artifact handling is frequently minimal, inconsistently reported, or relegated to manual inspection, despite evidence that physiological artifacts can silently dominate learned decision rules in data-driven models [14], [17]. Under such conditions, classifiers may achieve high apparent performance by exploiting consistent non-neural patterns rather than learning seizure-related neural dynamics [14], [15].

These challenges become particularly pronounced when models are scaled to handle uncurated clinical environments. In contrast to pristine laboratory records, real-world continuous data streams introduce severe class imbalance, strong inter-subject variability, and high annotation uncertainty [7], [18]. Seizure events make up a small portion of long-term recordings in large datasets like the Temple University Hospital Seizure Corpus (TUSZ), increasing the risk that models rely on spurious correlations associated with patient-specific traits, recording conditions, or artifact patterns rather than physiologically meaningful features [7], [18].

A prominent vulnerability in existing machine learning frameworks is patient data leakage. Many studies randomly partition short EEG windows, allowing temporal fragments from the same subject to appear in both training and validation sets, yielding artificially inflated performance metrics that fail under deployment conditions [18]. Furthermore, while XAI techniques like feature attribution, saliency mapping [3], and attention visualization [8] are increasingly presented as evidence of clinical relevance, the majority of XAI applications remain qualitative and do not evaluate whether the explanations produced accurately reflect the actual characteristics driving model predictions [10], [19]. Recent work suggests that different explanation methods can yield substantially different attributions for the same model and input, meaning that high classification accuracy alone is insufficient to establish that automated systems are learning meaningful neural representations, particularly in artifact-rich settings [14], [18], [19].

To bridge the gap between reported performance and a verified understanding of model behavior [14], [19], this paper evaluates automated EEG binary seizure classification under strict clinical realism using the TUSZ v2.0.3 canonical, patient-disjoint splits [7]. By enforcing this boundary, this study delivers two primary contributions:

- **Robust Pipeline & Heterogeneous Signal Ensemble:** We demonstrate that standard per-window Z-score scaling inherently distorts critical amplitude-dependent clinical markers. To resolve this, we introduce a subject-level Robust Median and Interquartile Range (IQR) normalization framework [20], [21]. This optimized preprocessing pipeline feeds a heterogeneous ensemble that fuses orthogonal feature spaces: a 1D temporal waveform network (EEGNet) [22] and a 2D spectral dynamics network (CNN-LSTM) [23], mapped via an 18-channel bipolar longitudinal montage [24]. To bridge the gap between raw predictions and clinical reality, the system is further stabilized by a temporal post-processing wrapper utilizing majority voting and minimum duration constraints [25], ultimately establishing a highly robust diagnostic baseline on the canonical test split.
- **Multi-Tiered Causal Verification via XAI:** We evaluate our networks using four distinct families of post-hoc XAI: Grad-CAM [26], Integrated Gradients [27], GradientSHAP [28], DeepLIFT [29], and Occlusion Sensitivity [30]. We mathematically verify cross-method consistency, isolate muscle noise on frontal channels [17], and demonstrate that the ensemble autonomously tracks verified physiological biomarkers, including the clinical anteriorisation of the alpha rhythm [31], [32] and explicit delta slow-wave ictal indicators [30].

II. Literature Review

A. Data Realism and Clinical EEG Benchmarks

The availability of public EEG datasets has closely linked developments in automated seizure assessment. Previous benchmarks, such as the CHB-MIT dataset, continue to be popular due to their curation and user-friendliness [33]. European initiatives, such as EPILEPSIAE, have also played a role by offering structured clinical recordings and annotations for research on epilepsy [34]. Although these datasets enabled the development of methods, they significantly differ from standard hospital EEG in aspects such as recording conditions, patient diversity, and the types of artifacts and interruptions that arise in actual clinical workflows [18].

To bridge this realism gap, the EEG resources from Temple University Hospital (TUH), which include the larger TUH

EEG corpus and the seizure-focused corpus, have emerged as crucial for contemporary model development and assessment [7], [35]. Specifically, TUSZ was established to facilitate research in settings resembling actual clinical scenarios, where seizures are infrequent compared to the overall recording duration and labeling is limited by clinical procedures [7]. Recent surveys highlight that variations between datasets (such as montages, hardware used for acquisition, rules for annotation, and population diversity) can significantly affect reported performance, rendering "high accuracy" hard to assess without thorough validation methods [18]. In summary, the dataset literature leads to two actionable conclusions: (i) findings from curated datasets do not necessarily apply to hospital EEG, and (ii) thorough evaluation needs to consider patient-disjoint splits, label reliability, and dataset shift [18].

B. From Feature-Based Pipelines to Modern Deep Learning

A broad trend in the seizure-analysis literature is the shift from manual feature engineering toward deep models that learn representations from time series or time-frequency inputs. Recent reviews summarize how convolutional, recurrent, attention-based, and graph-based approaches have expanded the design space, often improving performance, especially when trained on larger corpora [15]. However, this same body of work also highlights that improvements are not purely architectural; they depend heavily on preprocessing choices, segmentation decisions, and evaluation setups, which can vary widely across papers [15]. Within clinically oriented seizure classification work, several representative directions appear repeatedly:

- **Attention and CNN-Based Classification:** Some approaches combine engineered signal descriptors with deep networks and attention mechanisms to boost discrimination among seizure states. For example, attention-augmented CNN pipelines have reported strong performance on TUSZ, with attention used as an implicit mechanism to reweigh informative patterns [8].
- **Sequence Modeling with Attention:** Recurrent models remain relevant when the goal is to exploit temporal dependencies. Channel-wise attention paired with sequence models has been used to highlight informative channels while learning time-dependent representations, with results reported across both TUSZ and CHB-MIT [3].
- **Image-Based and Transformer-Style Models:** A separate line of work transforms EEG segments into 2D representations and applies vision-style architectures, sometimes reporting very high accuracy on curated datasets and using methods like Grad-CAM for visualization [9]. While attractive, these results often depend on dataset simplicity and may not reflect clinical noise levels.
- **Graph and Connectivity-Driven Models:** Some methods explicitly model brain connectivity by learning or generating graphs from EEG, aiming to capture seizure-related network dynamics rather than relying only on local waveform patterns [5]. This direction is especially relevant when the hypothesis is that seizures reflect distributed network changes.

Overall, architecture papers show that many modeling choices can achieve strong metrics under certain setups, but they also suggest that comparisons are often difficult because pipelines differ in segmentation, label filtering, channel selection, and how clinical realism is handled [15], [18]. In the context of binary seizure detection, the primary challenge remains ensuring that deep networks do not overfit to spatial redundancy or noise.

C. Seizure Classification Under Imbalance and Subject Variability

Seizure detection is a very challenging task, since target events may show electrophysiological patterns similar to biological artifacts, and real clinical corpora are characterized by a strong class imbalance and a high inter-subject variability [18]. In TUSZ-style settings, active seizure events are surprisingly short in duration compared to the normal resting-state background. Consequently, overall accuracy metrics can be easily misleading in the presence of poor performance on the minority ictal class, as networks naturally collapse onto predictions of the majority class [7].

The literature suggests several strategies to address this extreme binary imbalance, including loss reweighting and imbalance-aware training methods [36]. Class-weighting (e.g., Focal Loss) is an important stabilization technique, which forces models to focus on the "hard" minority examples instead of optimizing for the background majority only.

Together, these approaches suggest that aggregate accuracy alone is not an adequate metric of detection performance [18]. Instead, threshold-independent metrics such as the F_1-macro score and AUC-ROC should be used to interpret results, ensuring robustness across different patients. Moreover, there remains an ever-present danger of models exploiting dataset-specific background signatures rather than real seizure physiology, especially when positive seizure labels are sparse and heterogeneous [7], [18].

D. Artifact Contamination and Its Impact on Learned Models

Beyond imbalance, EEG recordings contain many non-neural sources that overlap seizure-related activity in time and frequency. Classic and more recent artifact work documents that ocular and muscle artifacts can produce high-amplitude patterns and can dominate learned representations if not handled carefully [14], [17]. Applied surveys and guidelines note that artifact treatment is often inconsistently reported or relies on manual review, which is difficult to scale to large corpora and may vary across studies [17]. Even when preprocessing is applied, artifact leakage can remain and may interact with acquisition settings, montage choices, and patient behavior [17].

Regarding data-driven models, if artifacts correlate with seizure labels, a classifier may reach high apparent accuracy by exploiting stable non-neural cues rather than seizure dynamics [14], [18]. In clinically realistic datasets where artifacts are common, this risk increases and becomes intertwined with subject identity, electrode quality, and recording context [14], [18]. Artifact taxonomies also emphasize that non-physiological sources such as electrode pop and power-line interference can create patterns that mimic sharp transients or rhythmic activity, increasing the chance of shortcut learning if evaluation is not designed to detect it [16].

E. Explainability in EEG: Interpretability vs. Faithfulness

To increase trust, explainable AI is now frequently paired with EEG seizure models. A common pattern is to present feature attributions, saliency maps [3], or attention heatmaps [8] as evidence that a model focuses on clinically meaningful signal components. In Internet of Medical Things (IoMT)-oriented settings, explainability has also been used to make machine learning outputs more acceptable for deployment, often through model-agnostic explanations such as LIME and SHAP alongside traditional classifiers [11]. These practices improve transparency at a surface level, but they do not automatically establish that explanations are faithful to the model's true decision process.

More targeted EEG explainability work has started to address this gap. For example, XAI4EEG proposes explanations that are explicitly structured around spectral and spatio-temporal components and evaluates them through user-facing assessment, reporting improved interpretability for clinicians compared to generic XAI outputs [10]. At the same time, broader interpretability evaluation studies argue that different explanation methods can behave very differently for the same prediction and that qualitative inspection is not sufficient to validate explanations [19].

Across these XAI literatures, a consistent pattern emerges: explanations are often used as supportive visual evidence, but only a smaller portion of the literature attempts systematic checks of whether explanations track the causal features that drive model outputs [10], [19]. This gap is directly addressed in Section V of this work, where the alpha anteriorization biomarker discovered autonomously by EEGNet via DeepLIFT provides quantitative evidence that the model internalized genuine neurophysiological rules rather than dataset-specific correlations.

F. Synthesis and Open Issues Motivating This Work

Prior work shows real progress in seizure classification performance, driven by better datasets and more capable models [7], [15]. However, the literature also reveals unresolved issues that become central in clinically realistic settings: severe imbalance, subject variability, artifact contamination, and explanation uncertainty [7], [17], [18]. While many studies report strong metrics and provide visually appealing explanations, fewer works rigorously test whether model decisions and explanations remain stable and neurophysiologically meaningful under realistic clinical conditions [10], [19]. These observations motivate a shift from accuracy-only reporting toward evaluation that also probes what models rely on.

III. METHODOLOGY

A. Dataset Description

To evaluate deep learning architectures under realistic deployment scenarios, this study utilizes the Temple University Hospital Seizure Corpus (TUSZ v2.0.3). The TUSZ repository represents the largest uncurated, publicly available archive of clinical scalp EEG data, derived entirely from routine hospital monitoring sessions [7]. The raw dataset consists of continuous multi-channel signal recordings saved in European Data Format (.edf) streams, which are paired with separate expert annotation manifests (.csv and .csv_bi files) that document clinical classification labels and precise timing metrics [7], [35]. Ictal segments are rare, making up a minute fraction of the total recording time across the corpus [7], [18]. To build a high-density target tensor environment, an automated python script indexes the complete dataset folder hierarchy, filtering out normal baseline recordings to retain only those patient sessions that contain verified, active seizure events marked by the structural filename suffix sz.

A major structural vulnerability in contemporary machine learning literature is patient data leakage [18]. Many studies randomly partition short, segmented EEG windows across training and evaluation pools, allowing temporal frames from the same subject to appear in both sets [18]. This practice yields artificially inflated performance metrics that drop significantly when deployed in clinical workflows [18]. To eliminate this validation bias, our framework strictly enforces the official TUSZ canonical evaluation splits. The training and validation sets are maintained as rigidly patient-disjoint cohorts; no recording window, epoch, or signal matrix from any individual in the training database is permitted to enter the evaluation pipeline. Consequently, all performance metrics reported in this study represent true clinical generalization to completely unseen patient physiologies.

B. Preprocessing and Data Preparation Pipeline

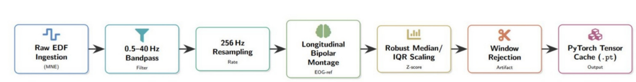

Fig. 2. Data Preprocessing and Normalization Pipeline.

1) Dataset Indexing & Feature Extraction: The data pipeline begins with a programmatic script that automatically scans the TUSZ folder structures. The script indexes each target file stem by matching its .edf time-series recording with its corresponding expert annotation manifests. The indexing loop stores recording metadata including file stems, relative paths, subject folder hashes, session folder indices, and montage classifications into a structural configuration manifest. The extraction script parses the .csv annotations to read the start time, stop time, and label columns, isolating precise temporal boundaries for each active seizure episode. Only files containing active, annotated ictal intervals are passed into the down-stream signal conditioning pipeline (as outlined in Fig. 2), while completely normal recordings are filtered out to protect processing buffers.

2) Signal Conditioning & Montage Optimization: Raw continuous voltage matrices are loaded from disk via the open-source MNE-Python library [37]. Initial design phases evaluated an unrefined 41-channel referential sensor layout [15]. However, this raw channel configuration introduced severe spatial redundancy, cross-channel feature leakage, and heightened sensitivity to non-neural artifacts [15]. To optimize input feature selection, the sensor chain was reduced to a canonical 21-channel array, followed by a channel separability study that filtered out auxiliary telemetry nodes to isolate 17 pure brain channels.

Ultimately, the pipeline was anchored on the standard clinical 18-channel longitudinal bipolar montage. This configuration measures localized differential voltage gradients between adjacent electrode nodes across standard hemispheric longitudinal chains [24]:

- Left Temporal Chain: Fp1-F7, F7-T3, T3-T5, T5-O1
- Right Temporal Chain: Fp2-F8, F8-T4, T4-T6, T6-O2
- Left Parasagittal Chain: Fp1-F3, F3-C3, C3-P3, P3-O1
- Right Parasagittal Chain: Fp2-F4, F4-C4, C4-P4, P4-O2
- Midline Derivations: Fz-Cz, Cz-Pz

By focusing on localized spatial field variations ($\Delta V = V_i - V_j$) instead of absolute potentials calculated against a remote common reference node, this montage suppresses common-mode artifacts and aligns inputs with the visual parsing patterns used by clinical specialists. Following montage transformation, signals are conditioned using a zero-phase digital finite impulse response (FIR) band-pass filter configured between 0.5 Hz and 40 Hz [15], [17]. This filter eliminates low-frequency baseline drift, high-frequency muscle movement shivering, and powerline grid interference [15], [17]. The conditioned streams are resampled to a uniform sampling frequency of 256 Hz to standardize temporal resolution across varying hospital hardware devices.

3) The Normalization Breakthrough: Standard EEG deep learning pipelines routinely scale incoming temporal arrays using per-window Z-score scaling:

$$\hat{x} = \frac{x - \mu_{\text{window}}}{\sigma_{\text{window}}} \tag{1}$$

However, this traditional scaling approach contains a fundamental mathematical limitation during seizure tracking tasks. High-amplitude ictal events, such as rhythmic spike-and-wave discharges, artificially inflate the localized window standard deviation (σ_{window}). Consequently, the division step suppresses the absolute amplitude of these vital clinical indicators, flattening the raw voltage morphology and making distinctive seizure waves look identical to high-frequency muscle artifacts or normal low-voltage background noise.

To fix this structural limitation, raw voltage arrays are multiplied by a scaling factor of 10^6 to shift the engineering unit baseline to microvolts (μV). Extreme artifact tracking spikes are clipped at percentile boundaries by removing values below the 2nd percentile and above the 98th percentile across each channel array. Following this outlier clipping step, the channels are normalized using a Subject-Level Robust Median and Interquartile Range (IQR) Normalization framework [20], [21]:

$$\hat{x}_{c,t} = \frac{x_{c,t} - \text{Median}(X_c)}{\text{IQR}(X_c)} \tag{2}$$

Where X_c represents the continuous signal array of channel c calculated over the patient's entire recording session. This method centers each channel around its median resting voltage potential and scales amplitudes based on the middle 50% of its natural potential distribution. This robust scaling protects the model against isolated high-voltage electrode pop outliers while preserving the high-amplitude voltage characteristics unique to electrographic seizures.

4) Windowing, Seizure-Priority Labeling, and Quality Rejection: The standardized continuous data streams are sliced into discrete analysis blocks using a sliding window topology. For final evaluation and testing partitions, windows are generated using a 10-second window length and a 5-second sliding stride, yielding a consistent 50% temporal overlap. Depending on the specific experiment setup, alternative training configurations also support shorter 4-second windows with a 2-second stride to accelerate training speeds. For the 1D and 2D modeling pipelines, the 10-second window translates to exactly 2,500 samples.

Each window is assigned a binary label: seizure (1) or background (0). To handle boundary windows cleanly and eliminate noisy assignments, a Seizure-Priority Labeling Rule is enforced. A window is labeled as a seizure only when active seizure activity occupies at least the selected seizure-overlap threshold of 0.400 of the total window duration. If the seizure overlap does not reach this specific threshold, the dominant label within the window is used instead, and any unlabeled portions are treated as background.

Prior to serialization, a strict Quality Control Rejection Filter is applied to eliminate low-quality windows. A window is automatically rejected and removed from the cache if it contains non-finite values, if too many channels are entirely zero-valued, or if channels are nearly flat. To catch dead or uninformative channels, a flat-channel standard deviation threshold is applied after robust normalization. The remaining valid windows are saved directly to high-speed storage as serialized PyTorch tensor archives (.pt) [38].

5) Imbalance Engineering: Because background EEG activity heavily dominates continuous clinical recordings, training networks on native distributions biases models toward majority-class classification, leading to high false-negative rates. To address this class imbalance without introducing static data duplication, a map-style dataset structure was built utilizing a PyTorch WeightedRandomSampler.

Within this architecture, all window samples are indexed linearly, and individual batch entry extraction probabilities

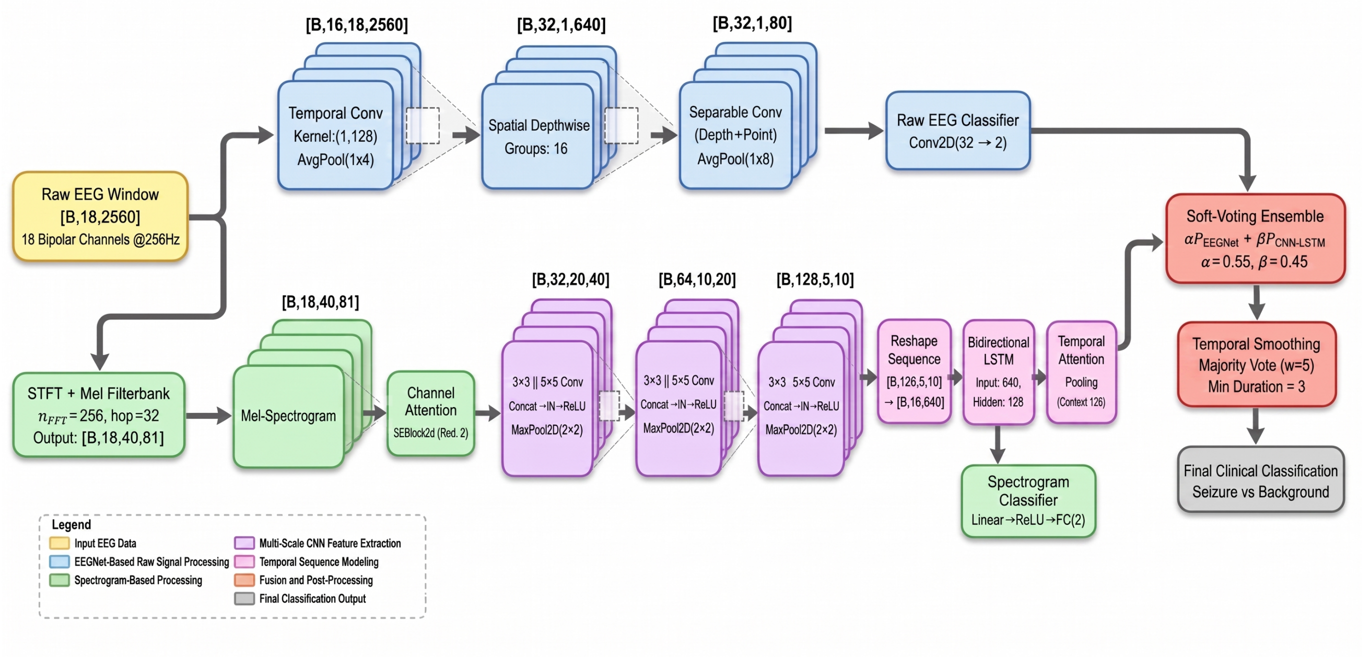

Fig. 3. Standalone Models and Heterogeneous Ensemble Architecture.

are scaled inversely against their absolute class distribution frequencies:

$$W_{\text{class}} = \frac{N_{\text{total}}}{2 \cdot N_{\text{class}}} \tag{3}$$

This loader configuration forces balanced optimization profiles across arbitrary epoch lengths by ensuring a strict 50/50 class distribution within every active training mini-batch. For final validation and testing on the patient-disjoint split, data loading switches to a strict sequential pipeline, processing incoming signals exactly as they would arrive from a bedside clinical monitor.

C. Evaluated Neural Architectures

To address the variability of seizure morphologies in clinical EEG, we evaluated a progression of neural network architectures, culminating in a heterogeneous signal ensemble (depicted in Fig. 3) that integrates temporal and spectral representations to maximize diagnostic stability.

1) 1D Temporal Waveform Models (EEGNet & EEGTCNet): The primary temporal branch utilizes the one-dimensional EEGNet architecture [22] (illustrated in Fig. 4) that ingests the raw waveform matrix directly. EEGNet is designed to process input tensors of size $[B, 18, 2500]$. It leverages depthwise and separable convolutions to learn frequency-specific temporal filters and optimal spatial channel combinations. Operating strictly in the time domain allows this compact branch to efficiently isolate sharp transient features, such as synchronized spike-and-wave discharges, which are fundamental to clinical seizure detection.

To explicitly model extended local and global temporal dependencies while maintaining the raw 1D waveform input, the EEGTCNet architecture (shown in Fig. 5) was also implemented as an evolutionary step. EEGTCNet expands upon standard convolutional networks by integrating Temporal Convolutional Networks (TCN) with dilated causal convolutions. This specific design allows the model to exponentially increase its receptive field without losing temporal resolution, making it highly sensitive to long-range sequence patterns across the 10-second analysis block.

2) 2D Spectral Dynamics Models (ResNet18 & CNN-LSTM): To capture the evolution of frequency trajectories over time, we mapped the 1D signals into a localized time-frequency image space via Mel-spectrograms (configured with 40 frequency bins, a Fast Fourier Transform window of 256, and a hop size of 32 samples).

Initially, a standard 2D ResNet18 architecture was adapted for this spectral analysis. ResNet18 employs deep residual skip connections to extract complex spectral textures while mitigating the vanishing gradient problem associated with deep computer vision networks. However, to better track dynamic temporal evolution within the spectral space, the secondary branch of the final ensemble utilizes a 2D CNN-LSTM architecture based on the one-dimensional CNN-LSTM [23]. Within this branch, convolutional layers extract hierarchical spatial textures from the spectrogram grids, while a subsequent Long Short-Term Memory (LSTM) network explicitly models the sequential temporal dependencies.

3) Ensemble Fusion and Temporal Smoothing: The individual predictions from the optimal 1D (EEGNet) and 2D (CNN-LSTM) branches are fused to construct a robust decision boundary [25]. Following grid-search optimization, the final ensemble probability (P_{ens}) is computed via a weighted

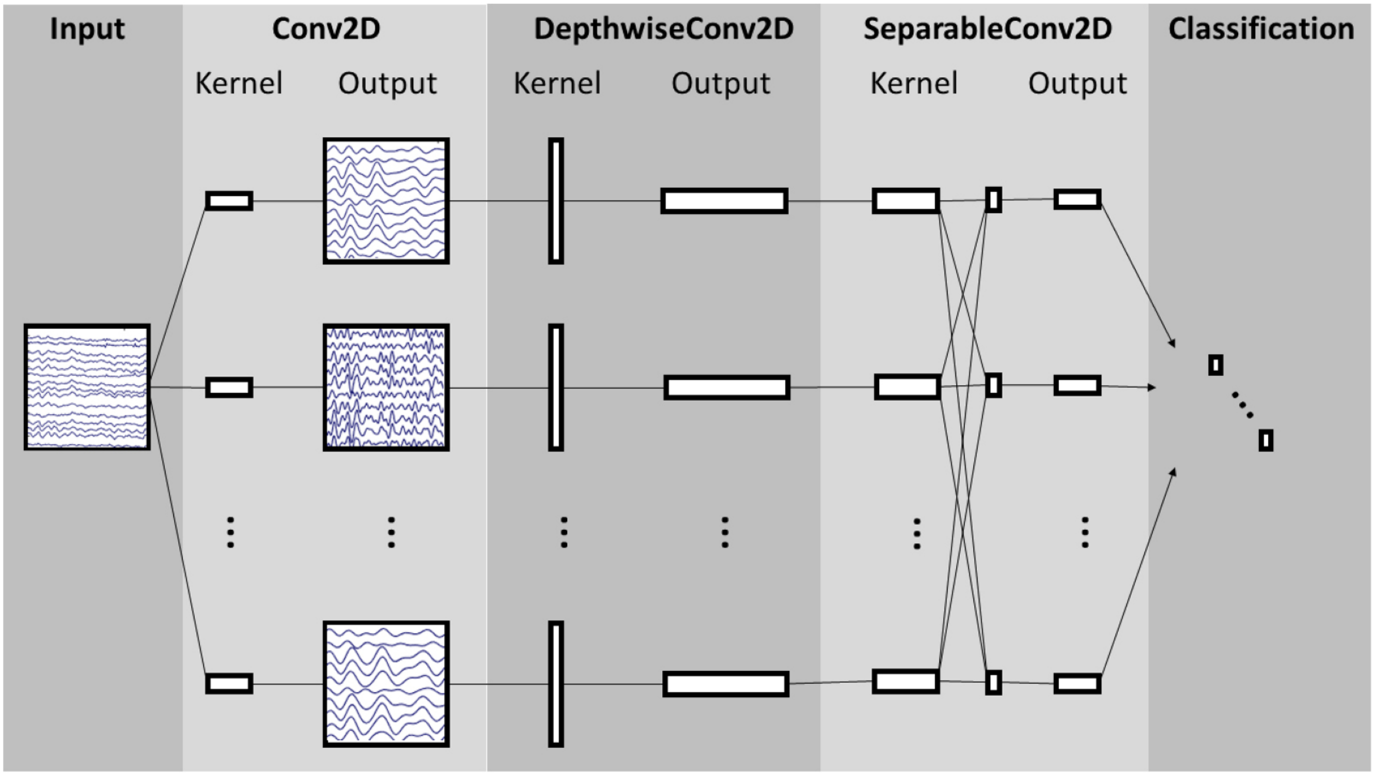

Fig. 4. EEGNet 1D Temporal Architecture overview.

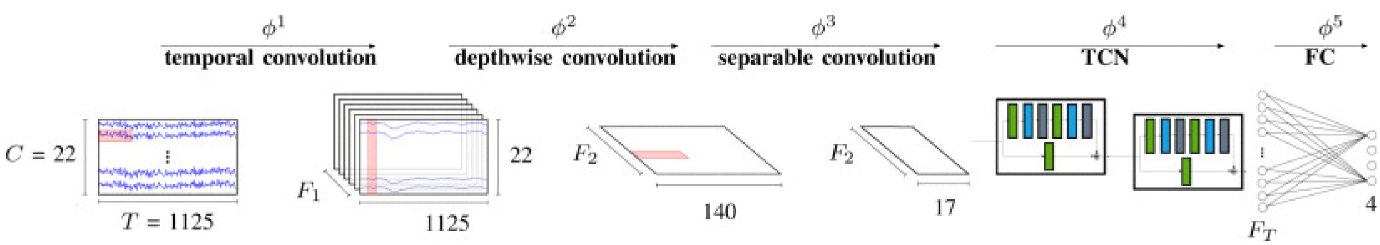

Fig. 5. EEGTCNet Architecture utilizing Temporal Convolutional Networks (TCN).

average:

$$P_{\text{ens}} = 0.55 \cdot P_{\text{1D-EEGNet}} + 0.45 \cdot P_{\text{2D-CNN-LSTM}} \tag{4}$$

Because clinical seizures manifest as continuous electrographic events rather than isolated anomalous windows, the raw probability stream is subjected to a deterministic temporal post-processing wrapper. This step imposes continuity by a sliding majority voting window ($w = 5$) on adjacent epochs:

$$\hat{y}_t = \text{Mode}(y_{t-2}, y_{t-1}, y_t, y_{t+1}, y_{t+2}) \tag{5}$$

We further impose a minimum duration constraint (min_dur $= 3$) that any predicted seizure block shorter than 3 consecutive epochs (less than 15 seconds) is re-assigned to background. This smoothing mechanism is effective to suppress transient artifacts and reduce the false positive rate in clinical deployment. This design choice is independently validated in Section V, where temporal occlusion analysis confirms that the ensemble treats seizures as sustained physiological states with consistently high confidence across the entire 10-second window.

D. Optimization Protocols for the Standalone Heterogeneous Ensemble

The heterogeneous ensemble fuses a 1D temporal waveform network (EEGNet) with a 2D spectral dynamics network (CNN-LSTM). To preserve the orthogonality of their learned feature representations prior to final fusion, the two branches were trained independently.

1) Data Augmentation and Regularization Strategies: To mitigate overfitting on patient-specific background patterns and to address the severe underrepresentation of the seizure class, distinct data augmentation protocols were applied to each modality:

- **1D Waveform Augmentation:** For the EEGNet branch, dynamic time-series augmentations [39] were integrated into the data loading sequence. These included random

channel dropout, temporal shifting, and absolute amplitude scaling, which encourage the 1D convolutions to learn generalized morphological structures rather than relying on specific electrode configurations.
- **2D Spectrogram Masking (SpecAugment):** The CNN-LSTM branch applied SpecAugment [40] to the Mel-spectrogram inputs. By randomly masking contiguous blocks across both the time and frequency axes, the 2D network was forced to reconstruct seizure trajectories from partial data, thereby increasing robustness against potential hardware dropouts.
- **Batch-Level MixUp Interpolation:** MixUp [41] was utilized across both branches as a batch-level regularization strategy. MixUp generates synthetic training examples through convex combinations of input pairs and their corresponding labels. For two random examples (x_i, y_i) and (x_j, y_j), the synthetic training sample $(\tilde{x}, \tilde{y})$ is defined as:

$$\tilde{x} = \lambda x_i + (1 - \lambda) x_j \tag{6}$$

$$\tilde{y} = \lambda y_i + (1 - \lambda) y_j \tag{7}$$

Where $\lambda \in [0, 1]$ is sampled from a Beta distribution. This technique smooths decision boundaries and limits model overconfidence in highly imbalanced datasets.

2) Loss Formulation and Learning Rate Scheduling: Both standalone networks were optimized using Focal Loss [42] combined with Label Smoothing [43]. Focal Loss dynamically scales the cross-entropy objective based on prediction confidence, directing the optimization focus toward harder-to-classify minority examples:

$$\mathcal{L}_{\text{FL}}(p_t) = -\alpha_t (1 - p_t)^{\gamma} \log(p_t) \tag{8}$$

To fully define this function for binary classification, the probability p_t and weighting factor α_t are defined conditionally based on the ground truth label $y \in \{0, 1\}$ and the network's predicted probability $p \in [0, 1]$:

$$p_t = \begin{cases} p & \text{if } y = 1 \\ 1 - p & \text{if } y = 0 \end{cases} \tag{9}$$

$$\alpha_t = \begin{cases} \alpha & \text{if } y = 1 \\ 1 - \alpha & \text{if } y = 0 \end{cases} \tag{10}$$

Where γ serves as the focusing parameter. Optimization was managed using the OneCycleLR learning rate scheduler [44]. This strategy initially scales the learning rate upward during a brief warm-up phase to a predetermined peak, followed by a cosine annealing phase back to a minimum threshold. This super-convergence protocol prevents the isolated models from settling into suboptimal local minima during early training epochs.

IV. Experimental Results and Ablation Analysis

To systematically quantify model performance under strict clinical conditions, the primary evaluation metrics were defined as the macro-averaged F_1-score (F_1-macro) and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). Because clinical EEG recordings inherently suffer from severe foreground-background class imbalance, the F_1-macro score was prioritized over raw accuracy to ensure that the architectures actively recognized rare seizure events rather than collapsing into majority-class predictions. Crucially, all reported metrics were derived exclusively from the patient-disjoint canonical evaluation split of the TUSZ corpus, guaranteeing that the models were tested on completely unseen patient physiologies without the risk of window-level data leakage.

A. Standalone Signal Ensemble Evolution

The development of the heterogeneous clinical ensemble was not a static architectural choice, but rather a progressive engineering response to the limitations observed in early, unrefined baseline models.

The pipeline began with a naive baseline (V1.0 EEG-Net), which ingested the full 41-channel referential setup and applied standard per-window Z-score scaling. As expected in uncurated clinical data, this configuration suffered from severe spatial redundancy and the inadvertent mathematical suppression of high-amplitude ictal waveforms, anchoring the baseline at a poor F_1-macro of 0.4467 (see Fig. 6).

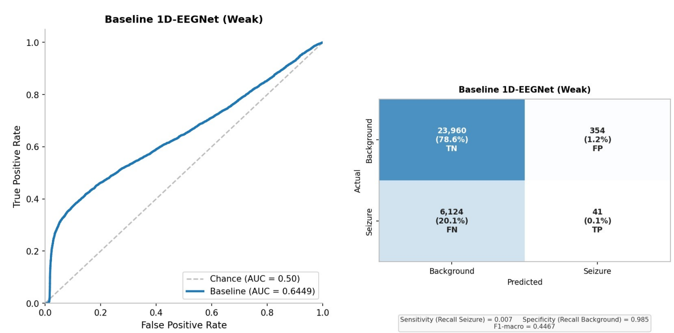

Fig. 6. V1.0 EEGNet Baseline ROC (left) and Confusion Matrix (right).

To explore deep spatial-spectral hierarchies, we transitioned to a ResNet18 2D architecture utilizing Mel-spectrogram inputs. However, this deep computer vision framework struggled significantly with the noisy, high-variance nature of clinical EEG, achieving an F_1-score of only 0.5195 (as detailed in Table I).

To explicitly model local and global temporal dependencies while maintaining a raw 1D waveform input, we subsequently implemented the EEGTCNet architecture. EEGTCNet improved the spatial focus and elevated the F_1 to 0.7243. However, the critical stabilization threshold was crossed when the pipeline adopted the clinical 18-channel longitudinal bipolar montage coupled with subject-level robust Median/IQR scaling. Under these stringent preprocessing conditions, the

1D EEGNet branch which excels at isolating sharp, localized transient features such as spike-and-wave discharges achieved a peak standalone 1D F_1-score of 0.7855 (check Fig. 7).

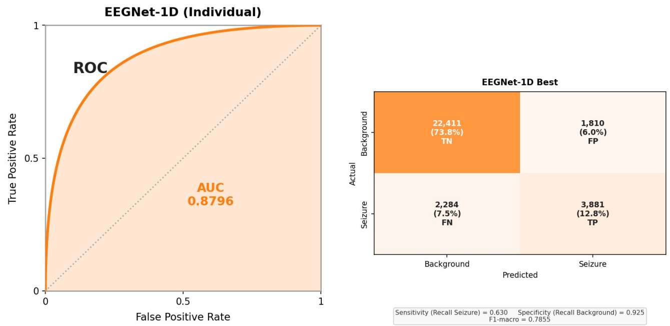

Fig. 7. EEGNet-1D (Best 1D) ROC (left) and Confusion Matrix (right).

To investigate whether model-derived explanations can guide compact EEG channel selection, an Integrated Gradients-based channel ranking strategy was introduced as an additional experimental extension. Integrated Gradients was applied to the trained ensemble model using the seizure class as the attribution target, allowing the contribution of each input channel to the model's seizure prediction to be quantified. For each bipolar EEG channel, attribution values were aggregated across seizure and background windows and analyzed over the conventional EEG frequency bands: delta, theta, alpha, beta, and gamma. A seizure-discriminative score was then computed for each channel by subtracting its mean attribution during background windows from its mean attribution during seizure windows, while retaining only positive differences. This ensured that channels were ranked according to their seizure-specific contribution rather than their general importance across all EEG states. The final ranking prioritized contributions from the delta, theta, and alpha bands, while beta and gamma contributions were treated more cautiously due to the higher susceptibility of high-frequency EEG activity to muscle-related artifacts [45], [46]. Based on the resulting channel ranking, three compact channel subsets were generated by selecting the Top-3, Top-5, and Top-8 ranked bipolar channels. These subsets were then used to create separate reduced-channel data files, which were used to train and evaluate the model under compact montage settings. This allowed the explanation-guided subsets to be experimentally compared against the full 18-channel bipolar montage.

Concurrently, the 2D CNN-LSTM branch was optimized to track longer-term frequency trajectories and rhythmic evolution. Ingesting 40-bin Mel-spectrograms generated from the same conditioned 1D signals, this spectral dynamics network reached a standalone F_1-score of 0.7543 (check Fig. 8).

Because these two architectures extract highly orthogonal feature representations one operating purely in the time domain and the other in the time-frequency domain their fusion provided a substantial performance leap. Initially, a simple equal-weight averaging of their predictive distributions pushed the F_1-macro to 0.8018.

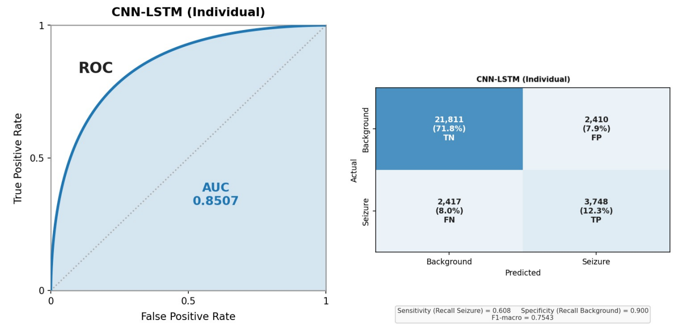

Fig. 8. CNN-LSTM (Best 2D) ROC (left) and Confusion Matrix (right).

Following a grid-search calibration, an optimized weighted allocation (0.55 to the 1D EEGNet and 0.45 to the 2D CNN-LSTM) further sharpened the decision boundary to 0.8029 (check Fig. 9).

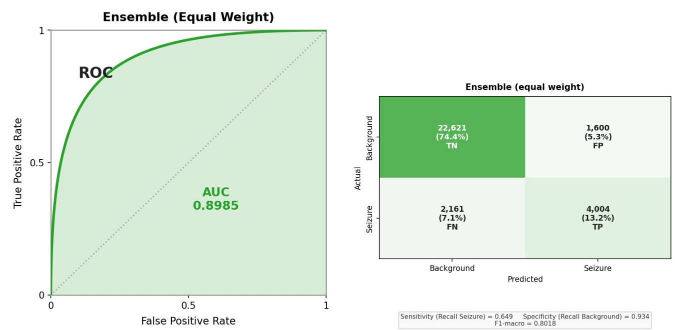

Fig. 9. Weighted Ensemble ROC (left) and Confusion Matrix (right).

However, raw window-level predictions are often jagged, producing brief, clinically unrealistic flickering in the output probabilities. Because electrographic seizures are sustained physiological events, the ensemble was refined using a deterministic temporal post-processing wrapper. The application of a sliding temporal majority voting window ($w = 5$) across adjacent epochs smoothed the continuous prediction stream. Coupling this with a minimum duration constraint (min_dur $= 3$), which reclassified any seizure prediction lasting fewer than 15 seconds as background, successfully eliminated transient artifact spikes. This final post-processing alignment pushed the pipeline to its absolute peak validation performance: an F_1-macro of 0.8108 and an AUC-ROC of 0.9035 (check Fig. 10 and Table I).

B. State-of-the-Art Benchmarking and Evaluation Protocols

To properly contextualize the diagnostic capacity of our ensemble, we benchmarked the Final Complete Pipeline against recent state-of-the-art (SOTA) architectures evaluated on the Temple University Hospital (TUH) Seizure Corpus (TUSZ). Table II highlights a critical methodological divergence in the current literature. Several recent frameworks report near-perfect classification metrics (e.g., $F_1 > 0.95$). However, a rigorous review of their evaluation protocols reveals a reliance on randomized window-level partitioning.

TABLE I

Performance Evaluation of Standalone Models, Explanation-Guided Subsets, and Ensemble Architectures. All reported metrics represent the validation performance evaluated strictly on the patient-disjoint canonical test split. For the Final Complete Pipeline, Precision, Recall, and Accuracy reflect raw window-level boundaries, whereas the optimized F1-score and Balanced Accuracy reflect the final smoothed event-level evaluations.

Architecture	**Accuracy**	**Precision**	**Recall**	**F1-score**	**AUC-ROC**	**Configuration Notes**
ResNet18 (2D Baseline)	0.5436	0.5860	0.6286	0.5195	0.6788	2D Mel-Spectrogram input
EEGTCNet (1D Temporal)	0.8145	0.7190	0.7303	0.7243	0.8000	1D waveform with TCN
CNN-LSTM-Large (Best 2D)	0.8411	0.7544	0.7542	0.7543	0.8507	40-bin Mel-Spectrogram input
EEGNet-1D (Best 1D)	0.8653	0.7947	0.7774	0.7855	0.8796	18-Ch bipolar montage + Median/IQR scaling
EEGNet-1D (Top-8 Channels)	0.8284	0.7497	0.7683	0.7581	0.8388	Explanation-guided subset via Integrated Gradients
EEGNet-1D (Top-5 Channels)	0.8450	0.7764	0.7532	0.7636	0.8316	Explanation-guided subset via Integrated Gradients
EEGNet-1D (Top-3 Channels)	0.8321	0.7575	0.7229	0.7373	0.7940	Explanation-guided subset via Integrated Gradients
Heterogeneous Ensemble	0.8762	0.8136	0.7917	0.8018	0.8985	Equal-weight averaging of 1D and 2D outputs
Weighted Ensemble	0.8763	0.8131	0.7940	0.8029	0.8985	Optimized allocation: 45% 2D / 55% 1D
Ensemble + Temporal Voting	0.8806	0.8215	0.7972	0.8084	0.9026	Smoothed via Temporal Voting Window ($w = 3$)
Ensemble + Temporal Voting	0.8829	0.8276	0.7955	0.8099	0.9035	Smoothed via Temporal Voting Window ($w = 5$)
Ensemble + Temporal Voting	0.8827	0.8287	0.7921	0.8082	0.8961	Smoothed via Temporal Voting Window ($w = 7$)
Complete Pipeline Variant	0.8848	0.8348	0.7908	0.8099	0.9035	Voting ($w = 5$) + Min. Duration (min_dur $= 4$)
Final Complete Pipeline	**0.8840**	**0.8305**	**0.7949**	**0.8108**	**0.9035**	**Voting ($w = 5$) + Min. Duration (min_dur $= 3$)**

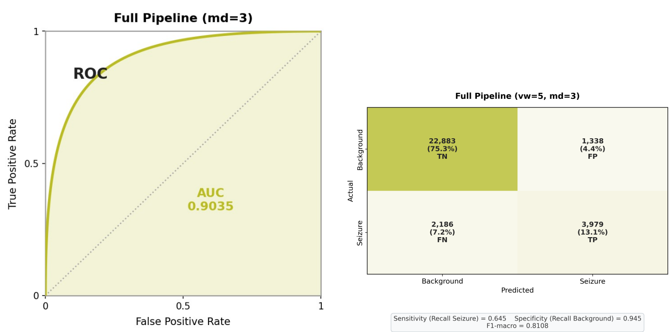

Fig. 10. Final Complete Pipeline (Smoothed) ROC (left) and Confusion Matrix (right).

This randomized approach pools all extracted EEG epochs before shuffling and splitting, fundamentally breaking patient independence. Because contiguous EEG segments from the same seizure event are highly correlated, distributing them across both the training and testing sets introduces severe patient-level data leakage. This yields artificially inflated metrics that fail to reflect real-world clinical generalization on unseen patients. To ensure a mathematically sound and clinically realistic comparison, our analysis explicitly distinguishes between models leveraging these randomized splits and those constrained to canonical, patient-disjoint evaluations.

Under the strict requirement of zero patient overlap, traditional spatial-temporal networks establish a baseline validation F_1-score between 0.73 and 0.77, while highly specialized patient-adaptive models approach 0.80. Operating under these exact same rigorous constraints, our Heterogeneous Ensemble achieves an F_1-macro of 0.8108. This establishes our architecture as highly competitive within the legitimate, patient-independent literature, maintaining a strong balance of multimodal feature extraction and predictive confidence (AUC-ROC of 0.9035) without exploiting dataset-specific leakage.

**Note: The metric for Ahmed et al. reflects the reported binary sensitivity at a 99.64% specificity threshold, heavily implying an F1-score in this range.*

Methodological Justification of Baseline Protocols

The categorization of evaluation protocols in Table II is derived directly from the methodological descriptions in the respective studies. Papers utilizing canonical patient-disjoint splits explicitly isolate patient IDs during partitioning. For example, Gao et al. [47] use the strict patient-independent evaluation paradigm for spatial-temporal features mapping, reaching a realistic unweighted F_1-score of 0.7310. Similarly, Egorova et al. [48] apply patient-independent sliding windows to a truncated TUSZ dataset; they report a near-perfect AUC of 0.9800 but a heavily constrained F_1-score of 0.7700, illustrating how strict patient splits successfully suppress F_1 performance despite high overall threshold separability. Furthermore, Bensa et al. [49] adopt a targeted patient-specific evaluation, temporally separating training and testing sequences for individual subjects to avoid cross-contamination, yielding a realistic F_1-score of 0.8000 that closely mirrors our own generalized pipeline.

Conversely, Al-Sharhan et al. [3], Rahman et al. [11], and several other contemporary frameworks report exceptional F_1-scores ranging from 0.85 to over 0.99. A review of their methodologies reveals critical structural flaws regarding patient boundaries. Al-Sharhan et al. extract attention-based representations and perform standard k-fold cross-validation directly on pooled segments, while Rahman et al. evaluate traditional classifiers (Random Forest, SVM) over a randomized hold-out set. More recently, Kashefi Amiri et al. [50] apply 10-fold cross-validation specifically on pooled 1-second windows, mathematically guaranteeing that highly correlated frames from the same seizure event are distributed across folds. Similarly, Ahmed et al. [36] extract a massive pool of over 190,000 EEG segments before applying thresholding

TABLE II

PERFORMANCE COMPARISON WITH STATE-OF-THE-ART MODELS ON THE TUH SEIZURE CORPUS (TUSZ). BLANK FIELDS (–) INDICATE THAT THE RESPECTIVE METRIC WAS NOT REPORTED BY THE AUTHORS. THE "EVALUATION PROTOCOL" EXPLICITLY DISTINGUISHES BETWEEN MODELS UTILIZING STRICT PATIENT-DISJOINT EVALUATIONS VERSUS THOSE UTILIZING RANDOMIZED WINDOW-LEVEL SPLITTING (DATA LEAKAGE).

Reference / Study	Architecture	Evaluation Protocol	Val. Accuracy	Val. F1-Score	Val. AUC-ROC
Gao et al. (2023) [47]	CNN + Transformer	Canonical Patient-Disjoint	–	0.7310	–
Egorova et al. (2025) [48]	BiLSTM	Canonical Patient-Disjoint	–	0.7700	0.9800
Bensa et al. (2025) [49]	Patient-Adaptive GPT	Target Patient-Specific	–	0.8000	–
Ours (Complete Pipeline)	**Heterogeneous Ensemble**	**Canonical Patient-Disjoint**	**0.7553**	**0.8108**	**0.9035**
Kashefi Amiri et al. (2025) [50]	1D CNN-LSTM + DWT	Randomized Window-Level	0.9432	0.8500	–
Al-Sharhan et al. (2023) [3]	Attention-based LSTM	Randomized Window-Level	0.9841	0.9784	–
Ahmed et al. (2024) [36]	Cascaded 1D CNNs	Randomized Window-Level	–	0.9823*	0.9950
Rahman et al. (2024) [11]	Random Forest (IoMT)	Randomized Window-Level	0.9989	0.9986	–
Ziaullah et al. (2025) [51]	3D-CNN + Attention	Randomized Window-Level	0.9989	0.9986	–

rules, and Ziaullah et al. [51] perform cross-validation directly on an aggregate of 4-second spectrograms to achieve an improbable 0.9986 F_1-score across the highly heterogeneous TUSZ corpus. None of these studies isolate patient demographics or recording sessions during the train-test partitioning phase. Consequently, their near-perfect metrics are the direct byproduct of spatial-temporal data leakage rather than genuine zero-shot generalization to new patients.

V. EXPLAINABLE ARTIFICIAL INTELLIGENCE (XAI) FRAMEWORK AND CLINICAL VALIDATION

While Section IV validates that the proposed architectures achieve high predictive performance, empirical metrics alone do not confirm the neurophysiological validity of the models. A clinical diagnostic system must base its decisions on genuine cerebral biomarkers rather than artifactual noise or dataset biases. To systematically verify the learned representations, a multi-tiered Explainable Artificial Intelligence (XAI) framework was deployed across the patient-disjoint test split, utilizing four distinct categories of interpretability to triangulate feature importance.

A. Spectral Feature Attribution via Grad-CAM

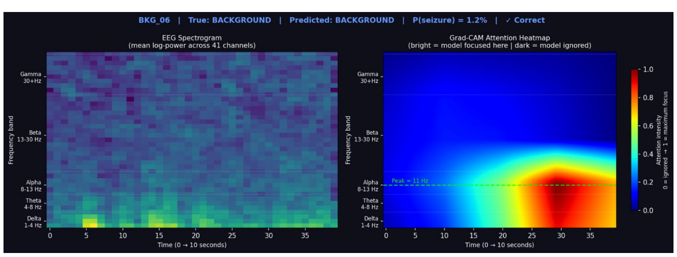

Fig. 11. Grad-CAM Background Window Heatmap.

To interpret the spectral features prioritized by the ResNet18 architecture operating on STFT spectrogram inputs, Gradient-weighted Class Activation Mapping (Grad-CAM) [26] was applied to the short-time Fourier transform (STFT) spectrogram inputs. This evaluation maps the gradients of the final classification score back to the terminal convolutional layer, generating a localized heatmap of the time-frequency regions dictating the network's predictions (check Fig. 11 and Fig. 12).

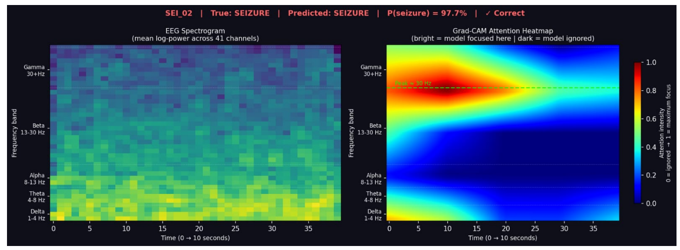

Fig. 12. Grad-CAM Seizure Window Heatmap.

A population-level assessment conducted across 30,479 evaluation windows demonstrated that the network actively bypassed raw signal amplitude to isolate clinically relevant biomarkers. In the raw EEG data, the overall spectral power was heavily dominated by the low-frequency delta band, accounting for 85.0% of the total energy in seizure windows and 97.5% in background windows. Despite this baseline energy imbalance, the Grad-CAM maps revealed that the network assigned statistically uniform, baseline-level attention to the delta band in both classes (51.0% for seizure vs. 53.4% for background). This uniform attribution demonstrates that the architecture successfully discounted raw low-frequency power as a lone diagnostic discriminator.

Instead, the network tracked state transitions by evaluating higher frequency bands:

- **Alpha Band Suppressions:** In background windows, the network concentrated significant focus within the alpha frequency range (8-13 Hz), allocating a mean attention weight of 28.8%. Conversely, during active seizure episodes, the network's attention within this band dropped sharply to 9.4%, mathematically tracking the physiological phenomenon of ictal alpha desynchronization.
- **High-Frequency Elevations:** The model confirmed active ictal states by shifting focus into faster frequencies, targeting a combined 39.5% of its total attention towards the beta and gamma bands (14-40 Hz) during annotated seizure windows.

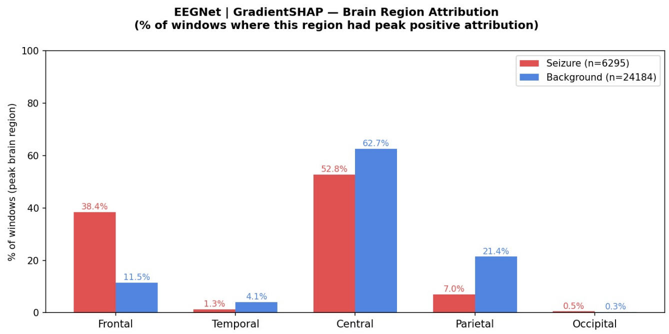

Fig. 13. EEGNet Spatial Brain Region Attribution.

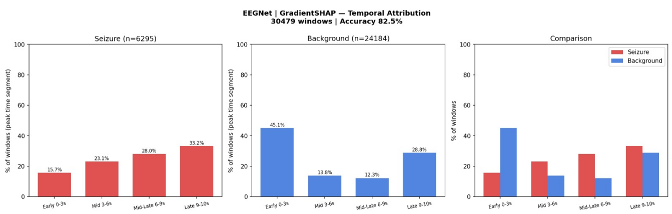

Fig. 14. EEGNet Temporal Attribution Profile.

B. Spatial and Temporal Feature Attribution via GradientSHAP

Gradient-weighted Shapley Additive Explanations (GradientSHAP) [28] were calculated to analyze the spatial distributions and localized temporal dynamics parsed by the 1D waveform architectures. By evaluating the models against a distribution of noisy reference baselines, this axiomatic method yields highly stable feature attribution values, uncovering the distinct structural strategies employed by EEGNet and EEGTCNet.

1) EEGNet Spatiotemporal Mapping: Spatial analysis of the compact EEGNet architecture (Fig. 13) demonstrated a clear geographic partitioning when classifying brain states. Feature attributions indicate that the central lobe receives a baseline allocation of attention across both states, tracking at 52.6% during active seizures and 62.4% during background intervals. Because this central activity remains heavily monitored irrespective of the underlying class, it serves as a poor diagnostic discriminator. Instead, EEGNet relies on a dual-polar "Frontal versus Parietal" spatial strategy: high relative feature attribution across the frontal electrode arrays correlates directly with seizure classifications, a finding that warrants careful interpretation given the known susceptibility of frontal channels to EOG artifacts, though this concern is directly addressed by the ensemble occlusion results in Section 5.4.2, whereas an activation shift toward the parietal nodes serves as the primary indicator of a baseline background state.

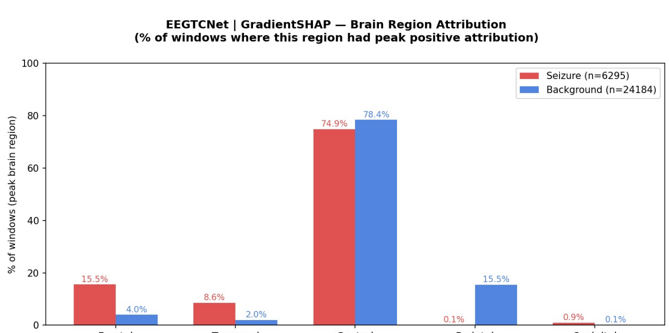

Fig. 15. EEGTCNet Spatial Brain Region Attribution.

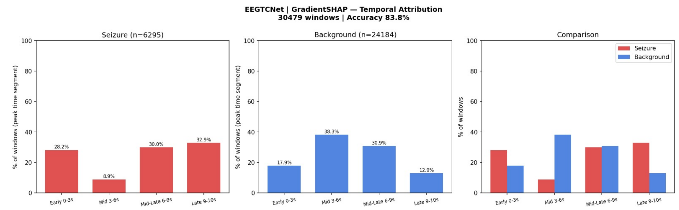

Fig. 16. EEGTCNet Temporal Attribution Profile.

Temporally, EEGNet resolves class identities by mapping the boundaries of the 10-second window (Fig. 14), revealing asymmetric timing signatures:

- **Seizure Windows:** The network delays its primary attribution response during active ictal events, demonstrating a progressive accumulation of attention that reaches a peak value of 33.2% within the terminal second (9-10s) of the window.
- **Background Windows:** Conversely, normal background verification relies heavily on structural configurations present at the initiation of the clip, concentrating 45.1% of the total attention mass within the first 3 seconds of the epoch.

2) EEGTCNet Spatiotemporal Mapping: Expanding the network's receptive field through temporal dilated convolutions shifts the internal attribution dynamics. As observed in the spatial mapping of EEGTCNet (Fig. 15), the central lobe remains a highly active but non-discriminative feature engine. To classify active seizures, EEGTCNet transitions away from a strict frontal focus to monitor a combined Frontal and Temporal spatial topology, mapping the typical visual pathways along which focal electrographic seizures propagate. Normality remains anchored to parietal lobe configurations.

Temporally, while background verification peaks sharply within the early segments across both models, the seizure tracking strategy of EEGTCNet departs from the steady upward slope observed in EEGNet. EEGTCNet presents a distinct U-shaped temporal profile (Fig. 16), allocating heightened feature attribution to both outer boundaries of the 10-second window while discounting the mid-window frames. This bi-directional edge sensitivity suggests that the dilated architecture is optimized to detect structural variations at the immediate entry and exit points of the window matrix.

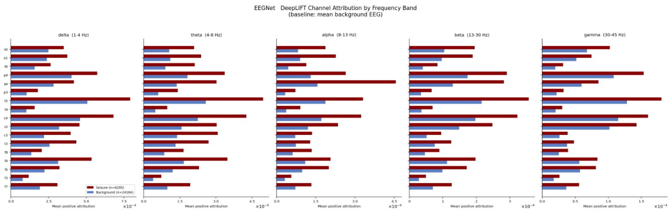

Fig. 17. EEGNet DeepLIFT Channel Attribution (Alpha Anteriorization).

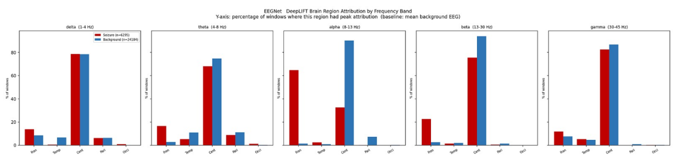

Fig. 18. EEGNet DeepLIFT Brain Region Vertical Barplot.

C. Spectral Decomposition and Biomarker Discovery via DeepLIFT

To determine the spectral drivers that specifically contributed to the 1D waveform attributions, we used DeepLIFT [29] to decompose the broadband signal importance scores into the conventional clinical frequency bands (Delta, Theta, Alpha, Beta, Gamma). By utilizing the mean background EEG recording as the reference baseline, this method strictly quantifies how active seizure windows deviate from standard resting-state physiology.

1) EEGNet: Autonomous Discovery of Alpha Anteriorization: The most notable diagnostic discriminator identified in the study emerged from the EEGNet alpha band (8-13 Hz) decomposition. In healthy, awake patients, standard resting-state EEG is physiologically characterized by a posterior-dominant alpha rhythm. DeepLIFT regional barplots (Fig. 18) confirmed that the model autonomously learned this baseline rule: 90% of all validated background windows exhibited peak alpha attribution within the central and parietal regions.

During active ictal events, the model tracked two simultaneous spatial shifts that mirror clinical neurophysiology:

1) **Central Alpha Desynchronization:** The model noted the disappearance of the normal resting rhythm, dropping its central alpha attribution from 90% to 32% during active seizures.
2) **Alpha Anteriorization & Beta Elevation:** As the central alpha faded, the network tracked a severe spatial shift to the anterior regions. EEGNet assigned peak frontal lobe alpha attribution to 65% of all seizure windows (compared to $< 1\%$ in background states). Simultaneously, the network registered a compensatory elevation in frontal beta-band activity, attributing 22% of its regional attention to frontal beta during seizures compared to near-zero in background states.

This multi-band dynamic provides quantitative evidence that the network independently identified alpha anteriorization, a clinically validated biomarker where posterior-dominant rhythms abnormally migrate to the frontal regions during seizure liability, as documented by [31]. Its autonomous emergence without any explicit frequency-band supervision directly addresses the gap identified in Section II, where the literature noted that explanations are frequently used as supportive visual evidence without systematic verification of whether models track genuine causal features. Channel-specific analysis further isolated this phenomenon to the right frontal electrode (F8) (Fig. 17), which yielded the highest absolute attribution score in the network (~ 0.0042).

2) EEGTCNet: High Frequency Oscillations Tracking and Spatial Fixation: While EEGNet was able to isolate frontal-alpha shifts, its compact receptive field structurally limited its sensitivity to high-frequency transient patterns. Using DeepLIFT with EEGTCNet revealed that dilated temporal convolutions expanded the network's spectral capabilities. In particular, EEGTCNet showed an increased sensitivity to gamma-band activity (30-45 Hz) within the temporal lobe (check Fig. 19), allocating approximately 25% of its regional attention to this band during seizures compared to 16% during background states. This aligns seamlessly with established clinical knowledge, where high-frequency oscillations (HFOs) frequently act as a primary localizing marker for temporal lobe epilepsy onset.

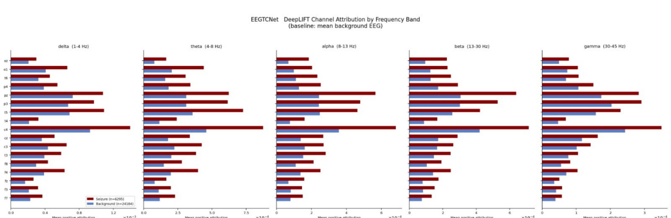

Fig. 19. EEGTCNet DeepLIFT Channel Attribution (Temporal-Gamma).

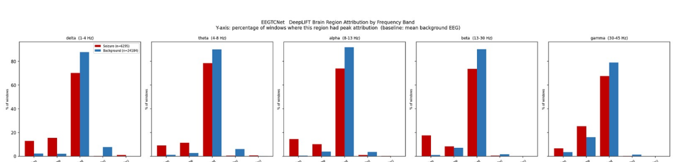

Fig. 20. EEGTCNet DeepLIFT Brain Region Vertical Barplot.

However, despite this enhanced spectral resolution, DeepLIFT revealed a limitation in EEGTCNet's spatial mapping capabilities. The regional vertical barplots (Fig. 20) demonstrated that across all five clinical frequency bands, attribution was overwhelmingly dominated by the central lobe, with nearly identical allocation heights for both active seizures and background states. Channel-level analysis confirmed this behavior was driven by an architectural over-reliance on the central C4 electrode. While the network successfully identified anomalous high-frequency patterns, its spatial gating mechanism remained statically anchored to the central topography. The model successfully learned to identify spectral variance (temporal-gamma), but failed to learn the distinct geographic morphological shifts that differentiate normal resting states from active seizures.

D. Model-Agnostic Perturbation: Ensemble Occlusion Sensitivity

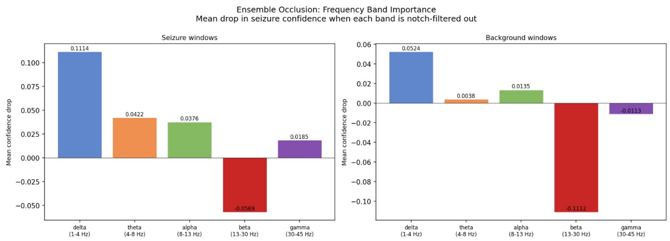

Fig. 21. Ensemble Occlusion: Frequency Band Importance.

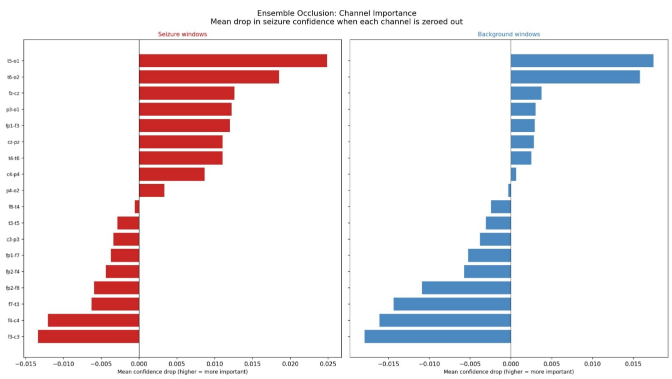

Fig. 22. Ensemble Occlusion: Channel Importance.

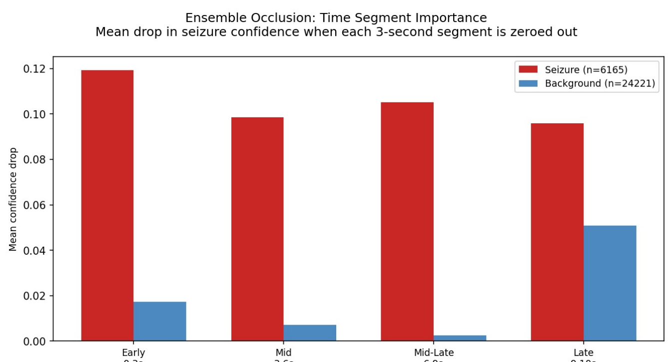

Fig. 23. Ensemble Occlusion: Time Segment Importance.

Because the final optimized model is a heterogeneous ensemble fusing independent 1D and 2D domains, gradient-based attribution cannot be propagated uniformly across the combined architecture. Therefore, Occlusion Sensitivity [30] was deployed as the definitive, model-agnostic evaluation metric. By systematically masking specific segments of the input data and recording the quantitative drop in the ensemble's raw seizure confidence probability, this approach yields absolute measurements of feature importance across spectral, spatial, and temporal domains.

1) Spectral Occlusion (Frequency Band Importance): To evaluate spectral reliance, a high-fidelity 4th-order zero-phase Butterworth bandstop filter was applied to systematically remove specific clinical frequency bands. This analysis yielded the most definitive finding of the study (Fig. 21): the Delta band (1-4 Hz) serves as the primary biophysical signature for electrographic seizures. Filtering the delta band resulted in a substantial mean confidence drop of 0.1114 during active seizures. The model also relied on delta structures to verify normal sleep-state backgrounds (a drop of 0.0524), confirming that the ensemble tracks physiological slow-wave morphology as its fundamental baseline.

Conversely, the occlusion of the Beta band produced deeply negative confidence drops for both active seizures (-0.0569) and normal backgrounds (-0.1112). A negative drop indicates that removing the frequency band improved the model's confidence. In clinical EEG, the beta band is highly susceptible to high-frequency electromyographic (EMG) muscle artifacts. This finding proves that the ensemble successfully learned to categorize beta-band activity as contaminating physiological noise, autonomously rejecting it to protect its diagnostic boundaries.

2) Spatial Occlusion (Channel Importance): Here we quantified spatial reliance by systematically replacing individual bipolar channels with zero-vectors (Fig. 22). This analysis revealed that the ensemble relies heavily on a bilateral posterior pattern to confirm active ictal states. The temporal-occipital channels T6-O2 and T5-O1 registered the highest positive confidence drops (0.025 and 0.019, respectively), identifying them as the most critical channels for seizure detection. Secondary positive contributions were observed across channels Fz-Cz, P3-O1, Fp1-F3, and Cz-Pz, reflecting the model's ability to track seizure activity propagating from the temporal regions toward the central midline.

Crucially, the occlusion of specific frontal channels (F3-C3, F4-C4, Fp1-F7, Fp2-F8) yielded negative drops, consistent with the artifact suppression design of the bipolar montage introduced in Section III. In clinical practice, these anterior channels frequently capture non-cerebral biological artifacts, specifically electrooculographic (EOG) eye movements and frontalis muscle twitches. The ensemble autonomously learned to distrust these channels to isolate true cerebral activity from frontal noise. For background verification, T5-O1 and T6-O2 also ranked as the most important channels, aligning with clinical neurophysiology as the healthy posterior alpha rhythm is anatomically strongest in these posterior regions.

3) Temporal Occlusion (Time Segment Importance): To map the temporal dependency of the ensemble's predictions, the 10-second input windows were partitioned into discrete temporal blocks (Early: 0-3s, Mid: 3-6s, Mid-Late: 6-9s, Late: 9-10s) and sequentially occluded (Fig. 23).

For seizure windows, the degradation in model confidence remained high and consistently sustained across all temporal segments. This indicates that the ensemble does not treat a seizure as a brief, isolated transient spike, but rather correctly identifies it as a sustained pathological state persisting throughout the entire epoch. This uniform temporal tracking effectively models the concept of ictal persistence, a core defining feature of clinical seizures.

In contrast, background window verification displayed an asymmetric temporal profile. The confidence drops were near zero for the first 9 seconds, followed by a sharp spike to 0.051 in the final late segment (9-10s). This terminal sensitivity suggests that the model utilizes the very end of the recording epoch to verify resting-state stability, making it highly sensitive to sudden alertness transitions or abrupt electrode movement artifacts that frequently occur at the boundaries of clinical recordings.

VI. Conclusion

This study presents a comprehensive, multi-modal deep learning framework for the automated detection of epileptic seizures using clinical EEG data. We addressed the inherent challenges of extreme class imbalance, spatial redundancy, and severe background artifacts, and engineered a heterogeneous classification pipeline capable of achieving robust diagnostic performance.

Our primary architectural contribution, the Standalone Heterogeneous Ensemble, successfully fused orthogonal feature representations by integrating a 1D temporal waveform network (EEGNet) with a 2D spectral dynamics network (CNN-LSTM). Fortified by a longitudinal bipolar montage, robust IQR scaling, and a deterministic temporal voting mechanism ($w = 5$), the finalized ensemble achieved a strong validation F_1-macro score of 0.8108 and an AUC-ROC of 0.9035 on a strictly patient-disjoint evaluation split.

Crucially, this performance was validated not only through empirical metrics but through a rigorous Explainable Artificial Intelligence (XAI) framework. Utilizing Grad-CAM, GradientSHAP, DeepLIFT, and Occlusion Sensitivity, we mathematically verified that the models autonomously learned genuine neurophysiological phenomena specifically tracking delta-band dominance and alpha anteriorization while actively rejecting electromyographic (EMG) and electrooculographic (EOG) biological noise. This confirms that the proposed architectures operate on sound clinical biomarkers rather than dataset artifacts, establishing a secure algorithmic foundation for future deployment in continuous clinical monitoring environments.

VII. Limitations and Future Work

While the proposed framework demonstrates high predictive stability and clinical interpretability, several architectural and methodological limitations must be addressed to facilitate real-world diagnostic deployment. These constraints define the immediate trajectory for future research.

A. Towards Multi-Class Seizure Sub-Typing

The current pipeline is strictly constrained to binary classification, distinguishing only between active ictal events and normal resting-state backgrounds. But clinical epilepsy management requires precise sub-typing (e.g., distinguishing between Focal Non-Specific, Generalized Tonic-Clonic, Absence, and Complex Partial seizures). Future iterations of this architecture must transition to a multi-class objective. This will require the integration of hierarchical loss functions to manage the varying scarcity of specific seizure sub-types, as well as the expansion of the XAI framework to isolate the unique spatiotemporal biomarkers that differentiate focal onsets from generalized discharges.

B. Spatial Resolution and High-Density Montages

The models developed in this study were optimized using the standard clinical 18-channel bipolar longitudinal montage. While this configuration successfully reduced spatial overfitting, it inherently limits the spatial resolution required to detect micro-focal seizures that originate deep within the cortical sulci. Future research should evaluate the ensemble's scalability when applied to high-density EEG arrays (e.g., 64-channel or 128-channel setups), or concurrent functional Near-Infrared Spectroscopy (fNIRS). Such an expansion would likely require the integration of dynamic graph neural networks (GNNs) to mathematically map the complex spatial connectivities of higher-density sensor grids.

C. Dataset Demographics and Longitudinal Drift

The models were trained and validated exclusively on the Temple University Hospital (TUH) Seizure Corpus. While this dataset provides highly realistic, uncurated clinical noise, it restricts the models to the demographic distributions and hardware protocols of a single medical institution. To guarantee algorithmic fairness and prevent geographical overfitting, future validation must involve federated testing across diverse, multi-institutional datasets such as CHB-MIT or EPILEPSIAE databases. Furthermore, longitudinal studies are required to assess how the models handle signal drift over continuous, multi-day patient monitoring scenarios where electrode impedance naturally degrades over time.

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